# Julia Syntax Cheatsheet

Applied Optimization with Julia

Dr. Tobias Vlćek



## Variables and Basic Types

### Variable Declaration and Types

```
# Basic variable declaration
x = 1                    # Implicit typing
y::Int64 = 5             # Explicit type annotation

# Common types
num_int = 42             # Integer
num_float = 19.99        # Float
is_student = true        # Boolean
name = "Julia"           # String

# Check type
typeof(num_int)          # Returns Int64
typeof(num_float)        # Returns Float64
```

### String Interpolation

```
name = "Julia"
age = 30
# Basic interpolation
message = "I am $age years old"
# Complex interpolation
greeting = "Hello, my name is $name and I am $age years old"
```

### Type Conversion

```
# Convert between types
float_num = Float64(42)     # Int to Float
int_num = Int64(3.14)       # Float to Int
str_num = string(42)        # Number to String
```

## Key Points

- Variables are dynamic, types are not
- Use `typeof()` to check variable type

- String interpolation is powerful for formatted output

# Vectors, Matrices, and Tuples

## Vectors

```
# Create vectors
grades = [95, 87, 91, 78, 88]    # Numeric vector
names = ["Mike", "Yola", "Elio"] # String vector

# Vector operations
push!(grades, 82)       # Add element to end
pop!(grades)            # Remove last element
popfirst!(grades)       # Remove first element

# Vector indexing
first = grades[1]       # Access first element
subset = grades[1:3]    # Access first three elements
```

## Matrices

```
# Create matrices
matrix = [1 2 3; 4 5 6]    # 2x3 matrix
# Matrix operations
matrix[2,3] = 17           # Change specific element

# Matrix arithmetic
matrix1 = [2 2; 3 3]
matrix2 = [1 2; 3 4]
sum_matrix = matrix1 + matrix2      # Matrix addition
prod_matrix = matrix1 * matrix2     # Matrix multiplication
element_prod = matrix1 .* matrix2   # Element-wise multiplication

# Broadcasting
matrix .+ 10                # Add 10 to each element
```

## Tuples

```
# Create tuples (immutable)
person = ("Elio Smith", 18, "Hamburg")
rgb = (255, 0, 0)

# Tuple operations
```

```
name = person[1]           # Access first element
age, city = person[2:3]    # Multiple assignment
```

### Key Differences

- Vectors: Mutable, 1-dimensional, good for lists
- Matrices: Mutable, 2-dimensional, good for linear algebra
- Tuples: Immutable, fixed-size, good for grouping related constants

---

## Comparison and Logical Operators

### Basic Comparisons

```
# Comparison operators
x == y    # Equal to
x != y    # Not equal to
x < y     # Less than
x > y     # Greater than
x <= y    # Less than or equal to
x >= y    # Greater than or equal to

# Examples
password_correct = (input == "secret123")
is_adult = (age >= 18)
can_afford = (price <= budget)
```

### Logical Operators

```
# AND operator (&&)
can_buy = (age >= 18) && (money >= price)    # Both conditions must be true

# OR operator (||)
need_coat = (temp < 10) || is_raining        # At least one must be true

# NOT operator (!)
is_closed = !is_open                         # Inverts boolean value
```

### Chained Comparisons

```
# Instead of
x >= 0 && x <= 10    # Check if x is between 0 and 10
```

```
# You can write
0 <= x <= 10          # More natural syntax

# Real-world examples
normal_temp = 36.5 <= body_temp <= 37.5
work_hours = 9 <= current_hour < 17
```

### Key Points

- Comparisons return boolean values (`true` or `false`)
- `&&` requires all conditions to be true
- `||` requires at least one condition to be true
- `!` inverts a boolean value
- Chained comparisons make range checks more readable

---

## Loops and Iterations

### For Loops

```
# Basic for loop with range
for i in 1:3
    println(i)        # Prints 1, 2, 3
end

# Iterating over array
fruits = ["apple", "banana", "cherry"]
for fruit in fruits
    println(fruit)    # Prints each fruit
end

# For loop with break
for x in 1:10
    if x == 4
        break         # Exits loop when x is 4
    end
end

# For loop with conditions
for x in 1:10
    if x <= 2
        println(x)
    elseif x == 3
        println("Three!")
    else
```

```
        break
    end
end
```

## While Loops

```
# Basic while loop
number = 10
while number >= 5
    number -= 1      # Decrements until < 5
end

# Infinite loop with break
current = 0
while true
    current += 1
    if current == 5
        break        # Exits when condition met
    end
end

# While loop with condition
lives = 3
while lives > 0
    lives -= 1       # Continues until lives = 0
end
```

## Nested Loops

```
# Nested loop example
sizes = ["S", "M", "L"]
colors = ["Red", "Blue"]
for size in sizes
    for color in colors
        println("$color $size")
    end
end

# Matrix iteration
for i in 1:3
    for j in 1:2
        println("Position: $i,$j")
    end
end
```

### List Comprehensions

```
# Basic list comprehension
squares = [n^2 for n in 1:5]    # [1,4,9,16,25]

# With condition
evens = [n for n in 1:10 if n % 2 == 0]    # [2,4,6,8,10]

# Nested comprehension
matrix = [i*j for i in 1:3, j in 1:3]    # 3x3 multiplication table
```

### Key Points

- `for` loops are best when you know the number of iterations
- `while` loops are useful for unknown iteration counts
- Use `break` to exit loops early
- List comprehensions offer concise array creation
- Nested loops are useful for multi-dimensional iteration

---

## Dictionaries

### Basic Dictionary Operations

```
# Create a dictionary
student_ids = Dict(
    "Elio" => 1001,
    "Bob" => 1002,
    "Yola" => 1003
)

# Access values
id = student_ids["Elio"]        # Get value by key
student_ids["David"] = 1004     # Add new key-value pair
delete!(student_ids, "Bob")     # Remove entry

# Check key existence
if haskey(student_ids, "Eve")
    println(student_ids["Eve"])
end
```

### Advanced Operations

```
# Dictionary with array values
grades = Dict(
    "Elio" => [85, 92, 78],
    "Bob" => [76, 88, 94]
)

# Get all keys and values
names = keys(grades)          # Get all keys
scores = values(grades)       # Get all values

# Iterate over dictionary
for (student, grade_list) in grades
    avg = sum(grade_list) / length(grade_list)
    println("$student: $avg")
end
```

### Common Methods

```
# Dictionary methods
length(dict)            # Number of entries
empty!(dict)            # Remove all entries
get(dict, key, default)# Get value or default if key missing
merge(dict1, dict2)     # Combine two dictionaries
copy(dict)              # Create shallow copy
```

### Key Points

- Keys must be unique
- Values can be of any type (including arrays)
- Use `haskey()` to safely check for key existence
- Dictionaries are mutable (can be changed)
- Keys are accessed with square brackets `dict["key"]`

---

## Functions

### Basic Function Definition

```
# Basic function with explicit return
function say_hello(name)
    return "Hello, $(name)!"
end
```

```
# Function with implicit return
function multiply(a, b)
    a * b    # Last expression is automatically returned
end

# Conditional return
function do_something(a, b)
    if a > b
        return a * b
    else
        return a + b
    end
end
```

## Function Scope

```
# Local scope example
function bake_cake()
    secret_ingredient = "vanilla"    # Only exists inside function
    return secret_ingredient        # Must return to access outside
end

# Variables outside function not accessible inside
global_var = 10
function scope_example()
    # Can read global_var but can't modify it
    return global_var + 5
end
```

## Multiple Dispatch

```
# Generic operation for all types
function operation(a, b)
    "Generic operation for $(typeof(a)) and $(typeof(b))"
end

# Type-specific implementations
operation(a::Number, b::Number) = a + b        # For numbers
operation(a::String, b::String) = string(a, b) # For strings

# Usage examples
operation(10, 20)          # Returns 30
operation("Hello", "!")    # Returns "Hello!"
operation("Hi", 42)        # Uses generic operation
```

### Key Points

- Functions can have explicit or implicit returns
- Last expression is automatically returned if no `return` statement
- Variables inside functions are local by default
- Multiple dispatch allows different behavior based on argument types
- Use `return` for early exits or conditional

---

## Package Management

### Basic Package Operations

```
# Import package manager
import Pkg              # Access as Pkg.function()
using Pkg              # Import all exported names

# Add packages
Pkg.add("DataFrames")  # Add single package
Pkg.add(["Package1", "Package2"])  # Add multiple packages

# Update packages
Pkg.update()           # Update all packages
Pkg.update("DataFrames")  # Update specific package

# Remove packages
Pkg.rm("DataFrames")   # Remove package
```

### Package Usage

```
# Import packages
import DataFrames     # Access as DataFrames.function()
using DataFrames     # Import all exported names

# Check installed packages
Pkg.status()         # List all installed packages
```

### Environment Management

```
# Environment operations
Pkg.activate("new_environment")    # Create/activate environment
Pkg.activate()                     # Activate default environment

# Project files
```

```
# Project.toml    - Lists direct dependencies
# Manifest.toml   - Complete dependency graph
```

### Key Points

- Use `import` for namespace control, `using` for direct access
- Always update packages regularly with `Pkg.update()`
- Create separate environments for different projects
- Project.toml and Manifest.toml track dependencies
- Package manager commands typically run in REPL

---

# DataFrames

## Creating DataFrames

```
using DataFrames

# Basic DataFrame creation
df = DataFrame(
    Name = ["John", "Mike", "Frank"],
    Age = [28, 23, 37],
    Salary = [50000, 62000, 90000]
)

# Empty DataFrame with specified columns
df_empty = DataFrame(
    Name = String[],
    Age = Int[]
)
```

## Accessing and Modifying Data

```
# Access columns
ages = df.Age                  # Get Age column
first_name = df.Name[1]        # First name in Name column

# Modify values
df.Salary[1] = 59000          # Update John's salary
df.NewColumn = zeros(3)       # Add new column

# Access multiple columns
subset = df[:, [:Name, :Age]]  # Select specific columns
row = df[1, :]                 # Select first row
```

## Filtering Data

```
# Filter with boolean indexing
high_earners = df[df.Salary .> 60000, :]

# Using filter function
high_earners = filter(row -> row.Salary > 60000, df)

# Multiple conditions
senior_high_earners = df[(df.Age .> 30) .& (df.Salary .> 60000), :]
```

## Data Manipulation

```
# Sort DataFrame
sorted_df = sort(df, :Age)                # Sort by Age
sorted_df = sort(df, [:Age, :Salary])     # Sort by multiple columns

# Add calculated column
df.Bonus = [row.Age > 30 ? row.Salary * 0.1 : row.Salary * 0.05 for row in
eachrow(df)]

# Iterate over rows
for row in eachrow(df)
    println("$(row.Name): $(row.Age) years old")
end
```

## Key Functions

```
nrow(df)              # Number of rows
ncol(df)              # Number of columns
names(df)             # Column names
describe(df)          # Summary statistics
push!(df, row)        # Add new row
select(df, :Name)     # Select columns
```

## Key Points

- Column access with dot notation (df.column)
- Use eachrow() for row iteration
- Boolean indexing for filtering
- push! to add new rows
- Broadcasting with dot operators (.>, .+, etc.)

# File Input/Output

## DelimitedFiles Operations

```
using DelimitedFiles

# Write matrix to CSV
data = [1 2 3; 4 5 6]
writedlm("data.csv", data, ',')      # Write with comma delimiter
writedlm("data.txt", data, '\t')     # Write with tab delimiter

# Read delimited files
matrix = readdlm("data.csv", ',')    # Read CSV file
matrix = readdlm("data.txt", '\t')   # Read tab-delimited file
```

## CSV and DataFrame Operations

```
using CSV, DataFrames

# Write DataFrame to CSV
df = DataFrame(
    Name = ["John", "Alice"],
    Age = [25, 30]
)
CSV.write("data.csv", df)            # Basic write
CSV.write("data.csv", df,            # Write with options
    delim = ';',                     # Custom delimiter
    header = false                   # No header
)

# Read CSV to DataFrame
df = CSV.read("data.csv", DataFrame)           # Basic read
df = CSV.read("data.csv", DataFrame,           # Read with options
    delim = ';',                               # Custom delimiter
    header = ["Col1", "Col2"]                  # Custom headers
)
```

## File Path Management

```
# Get current directory
@__DIR__                             # Directory of current file
pwd()                                # Current working directory

# Path operations
path = joinpath(@__DIR__, "data")    # Join path components
```

```
mkdir(path)                          # Create directory
isfile(path)                         # Check if file exists
```

### Key Points

- Use DelimitedFiles for simple matrix I/O
- CSV package for advanced DataFrame I/O
- Always use `@__DIR__` for relative paths
- Check file existence before operations
- Consider using try-catch for file operations

---

## Plotting with Plots.jl

### Basic Plots

```
using Plots, StatsPlots

# Line plot
plot(x, y,
    title="Line Plot",
    xlabel="X Label",
    ylabel="Y Label",
    legend=false
)

# Scatter plot
scatter(x, y,
    title="Scatter Plot",
    marker=(:circle, 8)
)

# Bar plot
bar(categories, values,
    title="Bar Plot"
)

# Histogram
histogram(data,
    bins=30,
    title="Histogram"
)

# Box plot
boxplot(group, values,
```

```
    title="Box Plot"
)
```

## Plot Customization

```
# Customize plot appearance
plot(x, y,
    title="Custom Plot",
    line=(:dash, 2),       # Line style and width
    color=:red,            # Line color
    marker=(:circle, 8),   # Marker style and size
    label="Data Series"    # Legend label
)

# Multiple series
plot(x, y1, label="Series 1")
plot!(x, y2, label="Series 2")  # Add to existing plot
```

## Saving Plots

```
# Save plot to file
savefig(plot_name, "path/plot.png")  # Save as PNG
savefig(plot_name, "path/plot.pdf")  # Save as PDF
savefig(plot_name, "path/plot.svg")  # Save as SVG
```

## Common Options

```
# Plot options
plot(
    legend=true/false,     # Show/hide legend
    grid=true/false,       # Show/hide grid
    size=(width,height),   # Plot dimensions
    dpi=300               # Resolution
)

# Line styles
:solid, :dash, :dot

# Colors
:red, :blue, :green

# Markers
:circle, :square, :diamond
```

### Key Points

- Use plot() for new plots, plot!() to add to existing
- Customize with named arguments
- Save plots in various formats
- StatsPlots extends plotting capabilities
- Multiple series can share one plot

# Bibliography